

BRAM MODESTUS NAIBAHO

Full Stack Developer | IoT & Network Enthusiast | Cloud Computing Learner

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PROFILE

Diploma III Computer Technology student at the Del Institute of Technology with experience building web-based information systems (Laravel, React, Node.js, MySQL) and LoRa-based IoT monitoring systems. Leading a three-person team as Project Manager & System Analyst in the LoRa Buoy final project for Lake Toba water quality monitoring (ESP32, Raspberry Pi, InfluxDB, Grafana), and continuing it to the second generation final project. Also developing a village information system (PKM) and a restaurant website with online ordering. Has a strong foundation in networking and cybersecurity (HCIA-Datcom, Cisco Ethical Hacker). Open to internships, full-time positions, and freelance projects in the field of web development and IoT.

MAIN PROJECTS & RESEARCH

LoRa Buoy-Based Lake Water Quality Monitoring System — Final Project (2025)

Project Manager & System Analyst · ESP32 · LoRa · Raspberry Pi 4 · MQTT · InfluxDB · Grafana

- Leading a three-person team developing a solar-powered buoy prototype that monitors four water parameters (temperature, pH, TDS, turbidity) in real-time with Agile methods, managing work schedules and budgets.
- Designing an end-to-end system architecture: ESP32 and LoRa on the buoy, LoRa receiver and Raspberry Pi 4 as gateway, MQTT, time-series storage in InfluxDB, and a Grafana dashboard running locally without relying on the internet.
- Analyze system requirements and select sensors (DS18B20, Analog pH, TDS, Turbidity), and compile complete technical documentation: requirements specifications, project planning, design, implementation, and testing.
- Tested the system in water for over 5 hours; data is displayed in Grafana in real-time with a delay of about 1–2 seconds and an aggregation of average, maximum, and minimum values per hour.

Second Generation Multi-Sensor LoRa-Based IoT Buoy — Final Project (2026)

LoRa · GPS · DO, pH, TDS, Temperature and Turbidity Sensors · Solar Power · Cloud IoT Platform

- Developing a second generation final project proposal for monitoring the water quality of Lake Toba: adding Dissolved Oxygen sensors (five parameters), GPS, and a cloud-based IoT platform with automatic warning notifications.
- Formulate development based on the results of the first generation evaluation, namely sensor calibration with standard fluids, LoRa transmission distance testing, and testing in open water.

Toba Ethnic Pizza Restaurant Information Website — Final Project

Team of 3 · WordPress · Elementor · MySQL · Apache · WhatsApp Ordering

- Build a restaurant menu information website that makes it easy for customers to access product information.
- Provide a digital ordering system that can be accessed online at any time to expand market reach.

Manalu Village Information System — Community Service (PKM)

WordPress · Elementor · MySQL · Apache · Manalu Village, Pakkat District, Humbang Hasundutan Regency

- Building a village information system with village info, village potential, announcements, news and village services features
- Helping the people of Manalu Village access village information digitally as part of the community service program.

SKILL

Programming Languages: Python, JavaScript, Java, PHP

Frontend: HTML, CSS, React

Backend: Node.js, Laravel

Database: MySQL, InfluxDB (time-series)

IoT & Embedded: ESP32, LoRa, Raspberry Pi, MQTT, Arduino IDE, water quality sensors (pH, temperature, TDS, turbidity, Dissolved Oxygen)

Monitoring & Design: Grafana, Fritzing, SolidWorks

Systems & Networks: Linux (Ubuntu, Kali), VMware, Server, Cloud

Security: Ethical Hacking, Cyber Security Fundamentals

Tools / DevOps: Docker, Git, GitHub

Project Management: Agile, scheduling and budgeting, technical documentation

EDUCATION and ORGANIZATION

D-III Computer Technology — Del Institute of Technology

2024 – Present

Active Member — Computer Technology Student Association (HIMATEK) IT Del

- Actively participate in the association's work programs, especially in developing members' technical skills.

IT Del Department of Arts and Culture

- Play a role in the development, preservation and promotion of arts and culture through organizing student creative activities and programs.

CERTIFICATION & AWARDS

- HCIA-Storage V5.0 — Huawei ICT Academy
- HCIA-Datacom V1.0 — Huawei ICT Academy
- Ethical Hacker — Cisco Networking Academy
- Introduction to Cyber Security — Cisco Networking Academy
- Data Analytics Essentials — Cisco Networking Academy
- Gemastik — Pusat Layanan Pembiayaan Pendidikan Kemendikbud Ristek
- Web Design Competition — PK-TECHSOFT / HIMA-RPL
- Aguna Course: Python, Linux, Network, dan Virtual Machine Fundamental